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Division: A

Semester: 7

Roll no:41

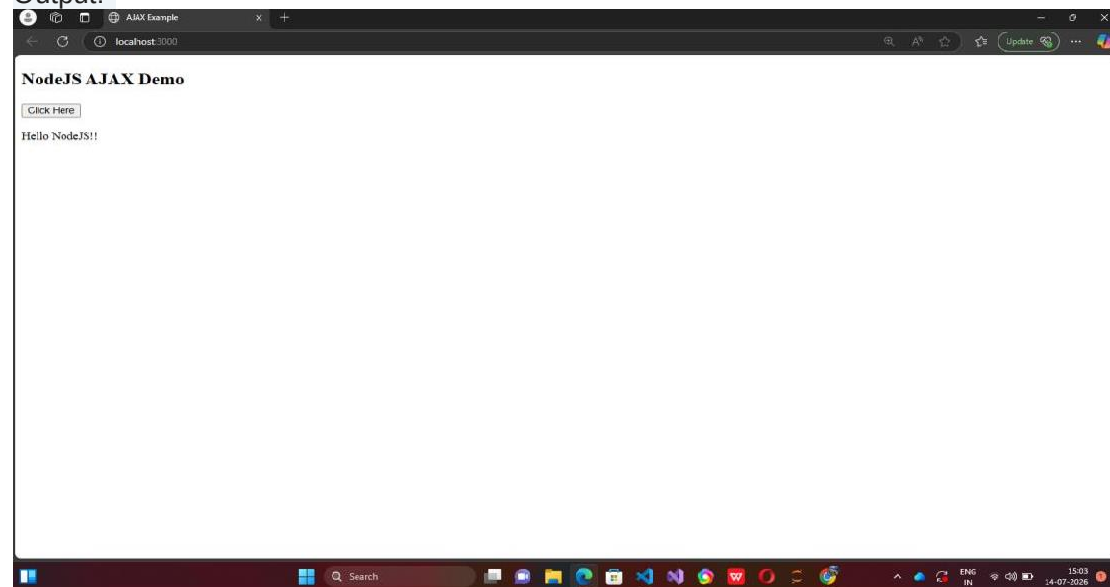
Sub: Node Js practical Assignment 1

Question 1:

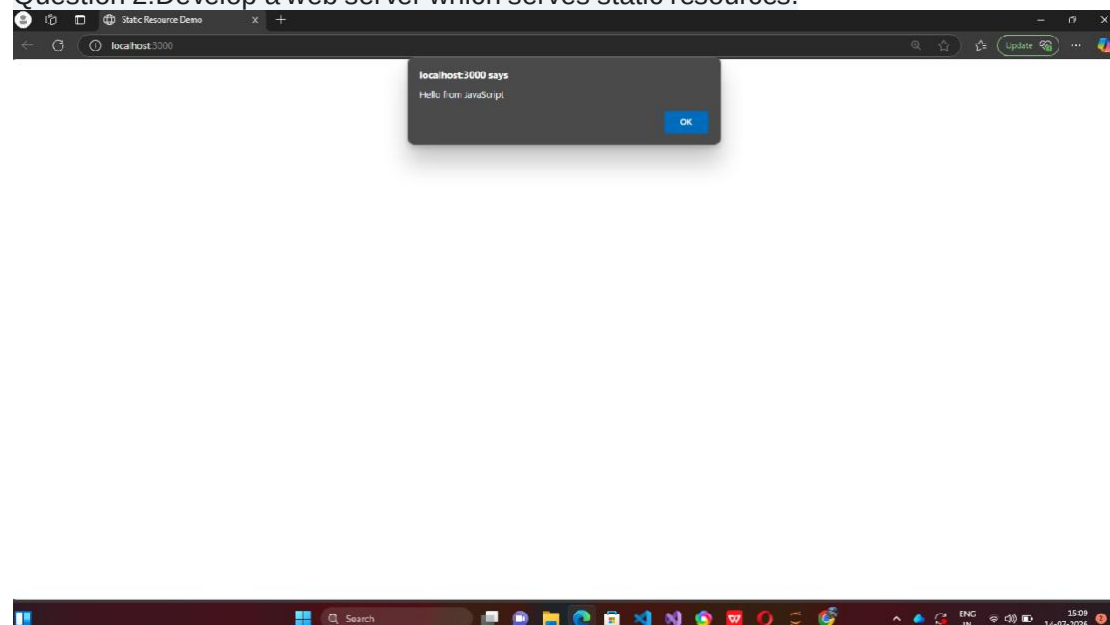
Develop nodejs application with following requirements:

- Develop a route "/gethello" with GET method. It displays "Hello NodeJS!!" as response.
- Make an HTML page and display.
- Call "/gethello" route from HTML page using AJAX call. (Any frontend AJAX call API can be used.)

Output:



Question 2: Develop a web server which serves static resources.



Question 3:

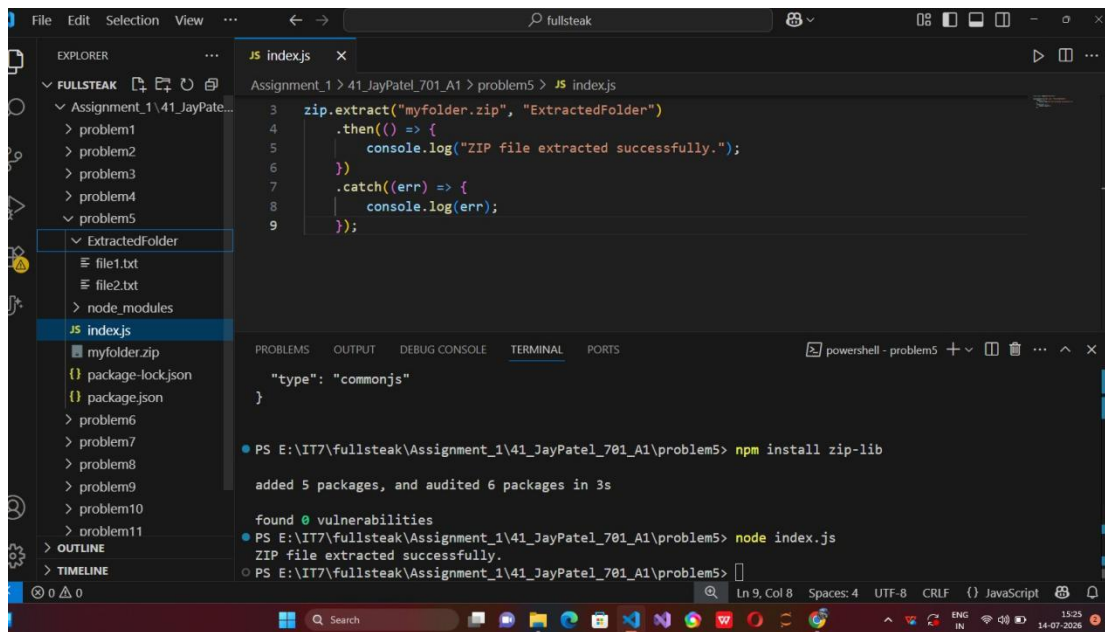
Develop a module for domain specific chatbot and use it in a command line application.

```
File Edit Selection View ... fullsteak
EXPLORER
FULLSTEAK
Assignment_1\41_JayPate...
  > problem1
  > problem2
  > problem3
  > problem4
  > problem5
  > problem6
  > problem7
  > problem8
  > problem9
  > problem10
  > problem11
  > problem12
  > OUTLINE
  > TIMELINE
JS index.js ...problem2 JS index.js ...problem3 X
Assignment_1 > 41_JayPatel_701_A1 > problem3 > JS index.js
9  rl.question("Ask something: ", (question) => {
10
11      console.log(getReply(question));
12
13      rl.close();
14
15  });
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - problem3
{
  "type": "commonjs"
}
PS E:\IT7\fullsteak\Assignment_1\41_JayPatel_701_A1\problem3> node index.js
Ask something: fees
Fees are Rs. 25,000 per semester.
PS E:\IT7\fullsteak\Assignment_1\41_JayPatel_701_A1\problem3> node index.js
Ask something: course
Our course is B.Sc IT.
PS E:\IT7\fullsteak\Assignment_1\41_JayPatel_701_A1\problem3>
Ln 15, Col 4 Spaces: 4 UTF-8 CRLF {} JavaScript 15:13 14-07-2026
```

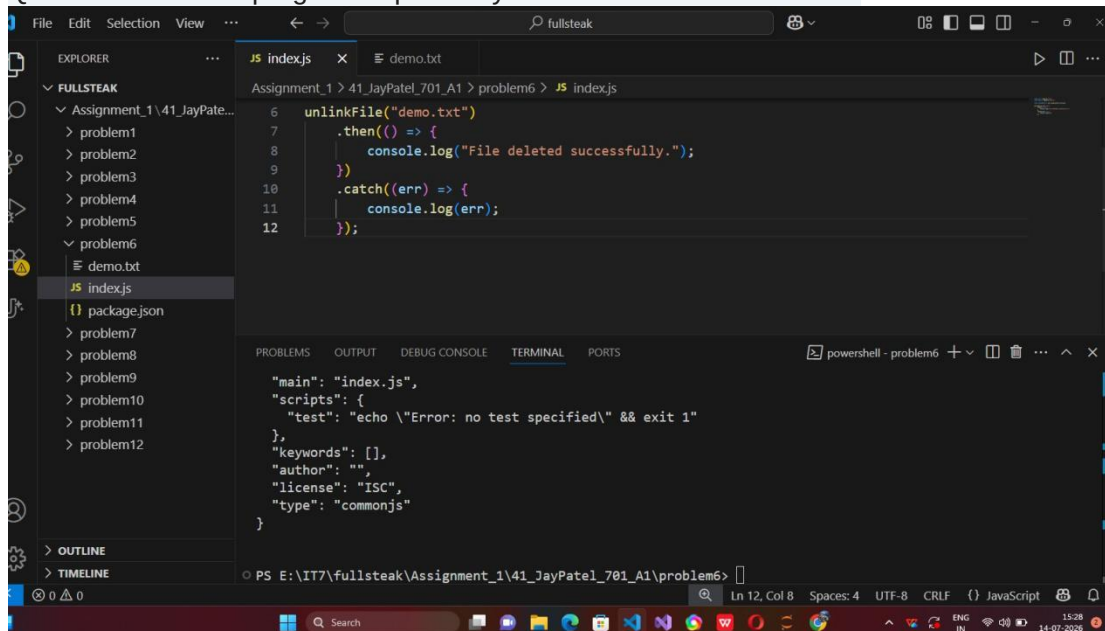
Question 4: Write a program to create a compressed zip file for a folder.

```
File Edit Selection View ... fullsteak
EXPLORER
FULLSTEAK
Assignment_1\41_JayPate...
  > problem1
  > problem2
  > problem3
  > problem4
  > myfolder
  > node_modules
  > JS index.js
  > myfolder.zip
  > package-lock.json
  > package.json
  > problem5
  > problem6
  > problem7
  > problem8
  > problem9
  > problem10
  > problem11
  > problem12
  > OUTLINE
  > TIMELINE
file1.txt file2.txt JS index.js X
Assignment_1 > 41_JayPatel_701_A1 > problem4 > JS index.js
3  zip.archiveFolder("myfolder", "myfolder.zip")
4
5  .then(() => {
6      console.log("ZIP file created successfully.");
7  })
8  .catch((err) => {
9      console.log(err);
10 })
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - problem4
PS E:\IT7\fullsteak> cd Assignment_1
PS E:\IT7\fullsteak\Assignment_1> cd 41_JayPatel_701_A1
PS E:\IT7\fullsteak\Assignment_1\41_JayPatel_701_A1> cd problem4
PS E:\IT7\fullsteak\Assignment_1\41_JayPatel_701_A1\problem4> node index.js
ZIP file created successfully.
PS E:\IT7\fullsteak\Assignment_1\41_JayPatel_701_A1\problem4>
Ln 9, Col 8 Spaces: 4 UTF-8 CRLF {} JavaScript 15:21 14-07-2026
```

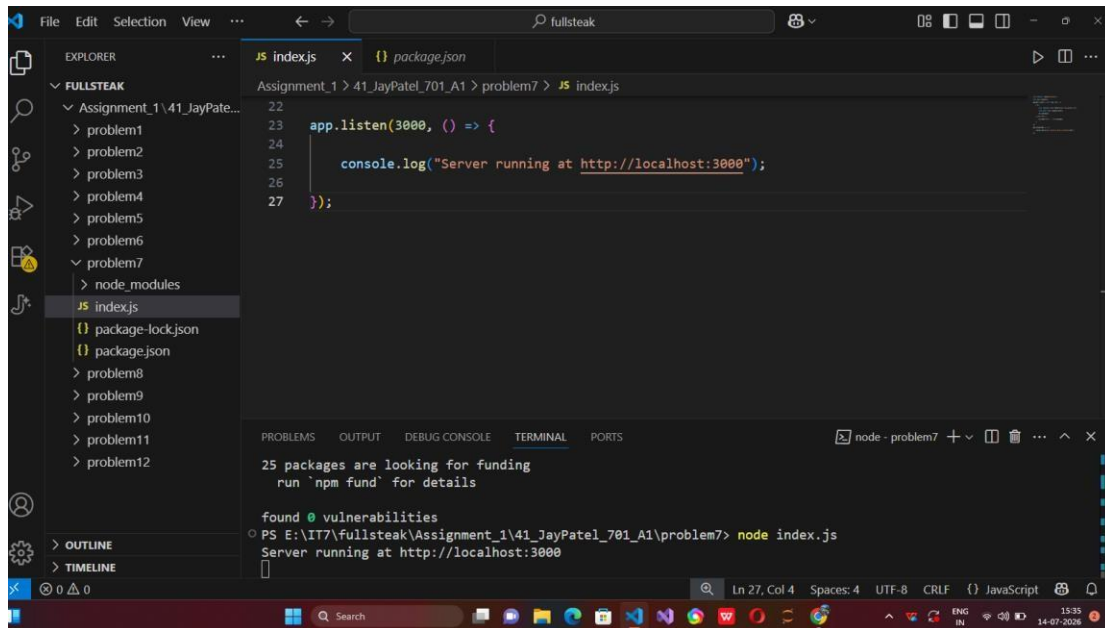
Question 5: Write a program to extract a zip file.



Question 6: Write a program to promisify fs.unlink function and call it.



Question 7: Develop a node web server with a route /google which can fetch data of google page using fetch() using async-await model and show as a response.

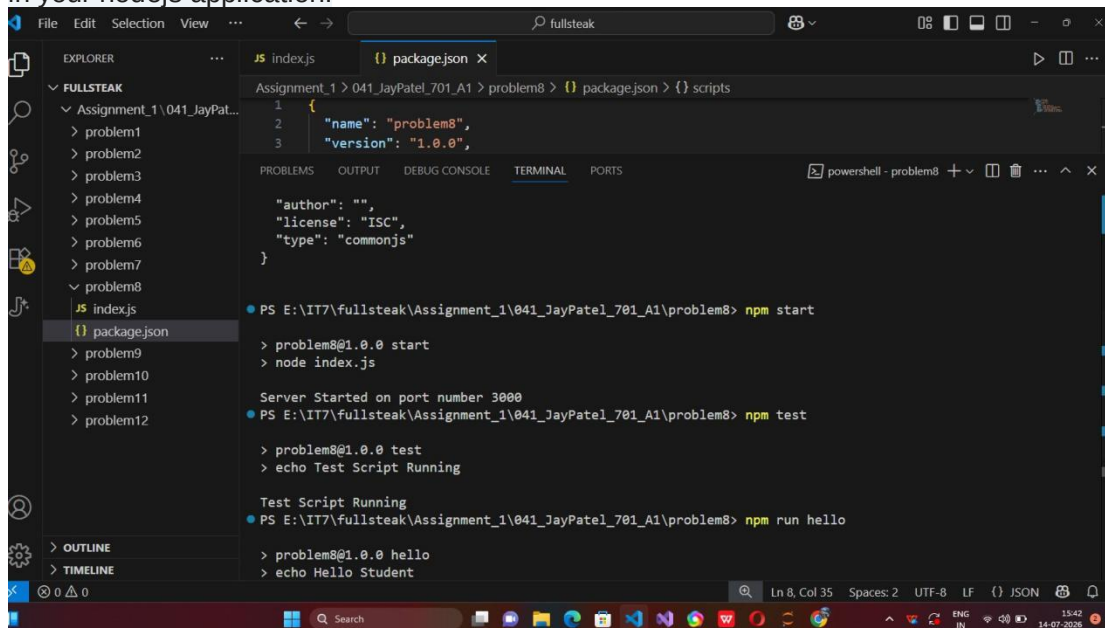


The screenshot shows a VS Code editor with a file explorer on the left. The file explorer shows a project named 'FULLSTEAK' with a folder 'Assignment_1\41_JayPatel_701_A1' containing subfolders 'problem1' through 'problem12'. The 'index.js' file is selected. The editor shows the following code in 'index.js':

```
22
23 app.listen(3000, () => {
24
25     console.log("Server running at http://localhost:3000");
26
27 });
```

The terminal at the bottom shows the command 'node index.js' and the output 'Server running at http://localhost:3000'.

Question8:Set a server script, a test script and 3 user defined scripts in package.json file in your nodejs application.



The screenshot shows a VS Code editor with a file explorer on the left. The file explorer shows a project named 'FULLSTEAK' with a folder 'Assignment_1\041_JayPatel_701_A1' containing subfolders 'problem1' through 'problem12'. The 'package.json' file is selected. The editor shows the following code in 'package.json':

```
1 {
2   "name": "problem8",
3   "version": "1.0.0",
4
5   "author": "",
6   "license": "ISC",
7   "type": "commonjs"
8 }
```

The terminal at the bottom shows the command 'npm start' and the output 'Server Started on port number 3000'. It also shows the command 'npm test' and the output 'Test Script Running'. Finally, it shows the command 'npm run hello' and the output 'Hello Student'.

Question 9:A program which calls useful functions in fs module.

The screenshot shows the VS Code interface with a file explorer on the left displaying a project structure under 'FULLSTEAK'. The file 'index.js' is selected. The code in 'index.js' is as follows:

```
4 fs.writeFile("demo.txt", "Hello NodeJS", (err) => {
9   fs.appendFile("demo.txt", "\nWelcome to FS Module", (err) => {
14    fs.readFile("demo.txt", "utf8", (err, data) => {
28      // ...
29    });
30  });
31 });
32 });
```

The terminal at the bottom shows the command `node index.js` being executed. The output is:

```
PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem9> node index.js
File created.
Data appended.

File Content:
Hello NodeJS
Welcome to FS Module

File renamed.
File deleted.
PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem9>
```

Question 10: A program which uses global objects in nodejs. Also read command line arguments and print it.

The screenshot shows the VS Code interface with a file explorer on the left displaying a project structure under 'FULLSTEAK'. The file 'index.js' is selected. The code in 'index.js' is as follows:

```
8
9 for (let i = 2; i < process.argv.length; i++) {
10   console.log(process.argv[i]);
11 }
```

The terminal at the bottom shows the command `node index.js Jay 41 surat` being executed. The output is:

```
Command Line Arguments:
Jay
22
Ahmedabad

This message is displayed after 2 seconds.
PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem10> ^C
PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem10> node index.js Jay 41 surat
Directory Name:
E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem10
File Name:
E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem10\index.js

Command Line Arguments:
Jay
41
surat

This message is displayed after 2 seconds.
PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem10>
```

Question 11: Develop a useful package and publish it on npmjs.com. Write a program to use it.

The screenshot shows a Visual Studio Code editor with a project named 'fullsteak'. The Explorer sidebar on the left shows a folder structure with 'Assignment_1' containing subfolders '041_JayPatel_701_A1' and 'problem11'. The file 'test.js' is selected. The main editor displays the code for 'test.js':

```
1 const calc = require("./cal");
2
3 console.log("Addition = " + calc.add(10, 5));
4 console.log("Subtraction = " + calc.sub(10, 5));
```

The TERMINAL panel at the bottom shows the output of running 'node test.js':

```
"keywords": [],
"author": "",
"license": "ISC",
"type": "commonjs"

PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem11> node test.js
Addition = 15
Subtraction = 5
PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem11>
```

Question 12: Write a node program having at least 3 dependent tasks and call using chained promises.

The screenshot shows a Visual Studio Code editor with the same 'fullsteak' project. The Explorer sidebar shows 'problem12' selected. The main editor displays the code for 'promise.js':

```
30 .then(task3)
31 .then(() => {
32 |   console.log("All Tasks Completed");
33 | });
```

The TERMINAL panel shows the output of running 'node promise.js':

```
"description": "",
"main": "index.js",
"scripts": {
  "test": "echo \\Error: no test specified\\ && exit 1"
},
"keywords": [],
"author": "",
"license": "ISC",
"type": "commonjs"

PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem12> node promise.js
Task 1 Completed
Task 2 Completed
Task 3 Completed
All Tasks Completed
PS E:\IT7\fullsteak\Assignment_1\041_JayPatel_701_A1\problem12>
```